House Price Prediction using Linear Regression

1. Introduction

Machine Learning enables computers to learn patterns from data and make predictions without being explicitly programmed. In this project, a Linear Regression model is developed to predict house prices using the California Housing Dataset available in Scikit-Learn.

The objective is to understand the complete machine learning workflow including data loading, preprocessing, exploratory data analysis, model training, evaluation, and reporting.

2. Dataset Description

The California Housing Dataset contains information collected from California districts.

Features used in the dataset:

- MedInc – Median Income
- HouseAge – House Age
- AveRooms – Average Number of Rooms
- AveBedrms – Average Number of Bedrooms
- Population – Population in the Area

- AveOccup – Average Occupancy
- Latitude – Geographic Latitude
- Longitude – Geographic Longitude

Target Variable:

- HousePrice (Median House Value)

The dataset contains 20,640 records and 8 input features.

3. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics of the dataset.

Missing Values Analysis

The dataset was checked for missing values. No missing values were found.

Statistical Summary

Descriptive statistics including mean, standard deviation, minimum, maximum, and quartiles were analyzed.

Correlation Analysis

A correlation heatmap was generated to identify relationships between variables.

Observations:

- Median Income showed the strongest positive correlation with house prices.
- Latitude and Longitude also influenced house prices.
- Average Bedrooms showed weak correlation with the target variable.

4. Model Development

Data Splitting

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

Algorithm Used

Linear Regression from Scikit-Learn was used.

The model attempts to fit a linear relationship between housing features and house prices.

5. Model Evaluation

Three evaluation metrics were used:

1. Mean Absolute Error (MAE)

Measures the average absolute prediction error.

2. Root Mean Squared Error (RMSE)

Measures prediction error while giving higher penalties to larger errors.

3. R-Squared (R^2)

Measures the proportion of variance explained by the model.

Typical results obtained are:

- MAE $\approx$ 0.53
- RMSE $\approx$ 0.74
- $R^2 \approx 0.57$

These results indicate that the model captures a significant portion of housing price variation.

6. Visualization

Actual vs Predicted Plot

The scatter plot compares actual prices with predicted prices.

The closer the points lie to the diagonal line, the better the model predictions.

Residual Plot

Residuals represent prediction errors.

A random distribution around zero indicates that the model assumptions are reasonably satisfied.

7. Conclusion

A Linear Regression model was successfully developed for house price prediction using the California Housing Dataset.

The project demonstrated the complete machine learning pipeline including data preprocessing, exploratory analysis, model training, evaluation, and visualization.

The model achieved satisfactory performance and can serve as a baseline for more advanced regression techniques.

8. Improvements Ideas

- Random Forest Regression
- Gradient Boosting
- Hyperparameter Tuning
- Feature Engineering
- Cross Validation

These approaches can improve prediction accuracy compared to simple Linear Regression.